SSU-Net: A Model for Mapping Burned Areas Using Landsat-8 and Sentinel-2

Shuai Cui, Jianyang Liu, Yuping Tian, Ying Quan, Bin Wang, Mingze Li, Yuan Zhang, Chengyuan Wang, and Yupeng Chang

***Abstract*— Information on burned areas (BAs) is particularly important for research on forest fires. Mapping BAs contributes to managing and preventing further destruction. BA mapping has long been constrained by the temporal and spatial resolution of satellite data and classification accuracy, while also neglecting the spectral characteristics of optical satellites. To address this issue, a novel method for mapping and extracting BAs using a semantic segmentation model has been proposed herein. Leveraging medium spatial resolution satellites, Landsat-8 and Sentinel-2, and after determining the optimal spectral band combination, a spectral–spatial module was integrated into the U-Net; this resulted in a new model, known as the SSU-Net. SSU-Net precisely extracted BAs from a single post-fire image. The optimal band combinations for Landsat-8 and Sentinel-2 were bands 7-5-4 and 12-8-4, respectively. SSU-Net achieved an $F1$ score of 0.9296 and an intersection over union (IoU) of 0.8707 on Landsat-8 data. It achieved slightly higher performance on Sentinel-2 with an $F1$ score of 0.9314 and IoU of 0.8752. Compared with U-Net, SSU-Net improved the $F1$ scores by 4.8% and 3.4% for Landsat-8 and Sentinel-2, respectively, and IoU by 7.7% and 5.9%. In terms of generalization, SSU-Net achieved an $F1$ score of 0.9151 when trained on Landsat-8 data and tested on Sentinel-2 data, and 0.9313 vice versa, with improvements of 2.1% and 5.1% over U-Net, respectively. SSU-Net also demonstrated higher accuracy and generalization in BA mapping, supporting the integration of multisource remote sensing data and emphasizing its crucial role in sustainable forest management on a global scale.**

***Index Terms*— Band selection, burned area (BA), forest fire, semantic segmentation, spectral features.**

I. Introduction

FORESTS are vital for maintaining water and soil balance [1], preserving biodiversity [2], absorbing carbon dioxide [3] and mitigating climate change [4]. However, extensive forest fires can cause forest degradation and diminish ecosystem services [5]. Accurate information on the boundary and location of burned areas (BAs) is particularly important for research on forest fires [6], [7], [8]. Traditional ground-based surveys are typically conducted for investigating BAs and assessing the scale and severity of forest fires [9]; however, they offer low efficiency and coverage and fail to meet the demand for extensive mapping of BAs. Remote sensing technologies have been used to precisely map BAs [10], [11], [12], [13]. Smaller BAs within regions cannot be accurately identified using products such as the MODIS monthly BA products, including the 500-m-resolution MCD64A1 [14] and 250-m-resolution FireCCI51 [15], so these areas are underestimated due to their coarse resolution. This shortcoming necessitates timely and accurate mapping of BAs at fine spatial and temporal resolutions. This will considerably help scientists, government agencies, and local authorities in assessing losses and formulating effective response and recovery strategies.

Landsat series satellites and Sentinel-2 constellations have been rapidly used for mapping BAs owing to their high spatial resolution and rich spectral information since their successful operation [16], [17]. Moreover, various algorithms have been developed based on Landsat and Sentinel-2 data, which can be broadly categorized into three types as follows:

1) *Spectral Index Methods:* Both Landsat and Sentinel-2 contain near-infrared (NIR) and short-wave infrared (SWIR) channels, which decrease and increase, respectively, in BAs [18]. Based on this principle, the normalized burn ratio (NBR) has been proposed for mapping BAs [19]. The theoretical range of NBR values is between -1 and 1, where negative values indicate BAs, 0 indicates unburned areas and positive values indicate vegetated areas. Additionally, indices such as the BA index (BAI) [20] and differenced NBR (dNBR) [21] have been developed using similar principles for mapping BAs. However, spectral index methods require prefire images, dense time-series data, and empirical thresholds, which incur significant time costs and limit future applications.
2) *Machine Learning:* Compared with spectral index methods, machine learning (ML) has garnered attention for its performance in BA extraction [25], including random forest [26], support vector machines [22], artificial

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The authors are with the Key Laboratory of Sustainable Forest Ecosystem Management, Ministry of Education, College of Forestry, Northeast Forestry University, Harbin, Heilongjiang 150040, China (e-mail: cs000514@nefu.edu.cn; ljy1530207445@nefu.edu.cn; tianyuping@nefu.edu.cn; quanying@nefu.edu.cn; wangbin@nefu.edu.cn; mingzelee@nefu.edu.cn; 2020220033@nefu.edu.cn; chengyuanwang@nefu.edu.cn; chanhyupeng@nefu.edu.cn).

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neural networks [28], and others [25]. ML methods depend on the distribution of training data and do not require assumptions; thus, they can achieve automated mapping of BA. However, these methods often exhibit poor generalization and transferability when applied to different landscapes and typically require retraining based on local conditions [29].

3) *Deep Learning (DL):* DL has made significant advances in remote sensing image object recognition [23] and segmentation [24]. This is because DL models can automatically extract more complex and abstract high-level features from data without requiring manual design [25]. They also have good generalization capabilities [26]. FCN [27], U-Net [28], and SegNet [29] are the major semantic segmentation models. Moreover, U-Net exhibited better performance on small datasets compared with FCN and SegNet, particularly in the segmentation of small objects and boundary details [30]. Therefore, U-Net is widely used for remote sensing image-based recognition and extraction of BAs and has yielded satisfactory results [31], [32], [33].

Knopp et al. [34] used uni-temporal Sentinel-2 imagery to propose an automatic BA mapping U-Net, achieving high accuracy but lacking generalization testing across other sensor data. Similarly, Brand and Manandhar designed a uni-temporal Sentinel-2-based U-Net tailored for mapping local BAS and leveraged the inherent generalization ability of U-Net to propose a globally applicable model with reliable accuracy [35]. Building on Knopp et al.'s study, Hu et al. [32] compared the applicability of four indices, three ML methods, and four DL models on uni-temporal Sentinel-2 imagery, demonstrating the superior classification accuracy of U-Net and its good performance with Landsat-8 data. Knopp et al. [34] identified an optimal band combination for DL models to map Sentinel-2 BA as B12-11-8, whereas Gascon et al. [36] replaced the B8 band with B8A because of its narrower spectral range within Sentinel-2. Considering the spectral similarities between Landsat-8 and Sentinel-2 [37], they recommended the band combination B7-6-5 for Landsat-8 [32]. Farasin et al. [38] simplified band selection based on spectral correlation coefficients among Sentinel-2 bands to enhance the accuracy of U-Net.

These researches show limitations in the remote sensing data sources, primarily emphasizing the applicability of Sentinel-2 for BA mapping and considering Landsat-8 only as a supplementary resource while overlooking Landsat-8's potential and its ability to generalize from Sentinel-2. By combining Landsat-8 and Sentinel-2, satellite revisit time can be reduced to 2.9 days [44], offering the potential for the near-real-time mapping of BAs using medium-resolution satellite imagery. Moreover, the optimal band combination for mapping BA with Landsat-8 and Sentinel-2 remains controversial. Current studies focusing solely on model perspectives clearly neglect the most crucial aspect of multispectral imagery, namely, the spectral characteristics of bands [44]. Therefore, it is necessary to determine the optimal band combination for mapping BA by considering both model and spectral aspects. Finally, this

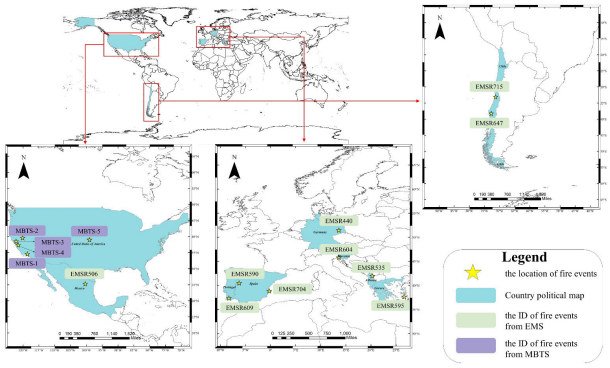


Fig. 1. Location map of 15 wildfire events. Blue areas represent the countries where the wildfires occurred. Yellow stars mark the locations of wildfires. Green rectangles contain the product ID information acquired from the EMSR. Purple rectangles contain the product ID acquired from the MTBS.

study aims to enhance U-Net's capability to map BA across these two data sources and improve its generalization ability between them. Specifically, the objectives of this study are as follows:

1) The spectral characteristics of multispectral imagery are considered to determine the optimal band combination of Landsat-8 and Sentinel-2 for identifying and mapping BA.
2) U-Net is enhanced by integrating a spectral–spatial module to improve model accuracy.
3) The enhanced U-Net model's generalization ability between Landsat-8 and Sentinel-2 data is qualitatively increased.

II. Study Area and Dataset

A. Study Area

This study primarily considered the spectral characteristics of multispectral images. To ensure the credibility and generalizability of the results, a total of 15 wildfire events occurring between 2020 and 2023 in Europe, North America, and South America were selected. The locations of these wildfire events represent various climate conditions (Köppen-Geiger classes: Cf, Cw, Cs, Bs, Ds, Df) [39], different elevations [40], and diverse forest structures [41]. Fig. 1 shows the geographic locations of all the wildfire events studied herein. Wildfire events were chosen using the Copernicus emergency management service (CEMS) rapid mapping–EMSR [37] and monitoring trends in burn severity (MTBS) projects [42]. Specifically, seven wildfires were selected from Europe, two from South America, and six from North America. The selection criteria aimed to include wildfires of varying sizes in regions with diverse climates, terrains, and vegetation types, thereby laying the groundwork for the generalizability of results. The conclusions drawn from these 15 wildfires served as a basis for future studies of similar events in the same regions. Table I provides detailed information about each wildfire event, and Section II-B2 provides extensive descriptions of EMSR and MTBS.

B. Dataset

1) Remote Sensing Data: Two types of remote sensing images (Landsat-8 and Sentinel-2) were used and applied to

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TABLE I
DETAILED DESCRIPTION OF THE REFERENCE BA MASK DATA AND FIRE IMAGERY DATA

Source	Number	Country	ID	Burned Area (ha)	Climate Type	L8 Time	S2 Time
EMSR	1	Germany	EMSR440	44.79	Cf	2020-06-03	2020-06-02
	2	Mexico	EMSR506	15610.74	Cw	2021-04-06	2021-03-30
	3	Albania	EMSR535	8011.68	Cs	2021-08-31	2021-08-14
	4	Spain	EMSR590	11926.6	Cs	2022-07-11	2022-07-15
	5	Greece	EMSR595	422.57	Cs	2022-07-30	2022-07-21
	6	Slovenia	EMSR604	3556.53	Cf	2022-08-05	2022-07-22
	7	Portugal	EMSR609	1368.92	Cs	2022-08-25	2022-07-78
	8	Chile	EMSR647	87337.09	Cs and Cf	2023-02-19	2023-02-07
	9	Spain	EMSR704	2005.44	Bs	2023-11-22	2023-11-21
	10	Chile	EMSR715	8782.58	Cs	2024-02-06	2024-02-12
MBTS	1	United States of America	CA3467911845220200812	12533.52	Cs	2020-08-20	2020-08-21
	2		CA3968112017120200814	18406.72	Ds	2020-09-03	2020-08-22
	3		CA3850412233720200817	127168.82	Cs	2020-08-25	2020-08-22
	4		CA3742412156820200816	164216.17	Cs	2020-09-03	2020-08-27
	5		KS3910009890020211215	49764.60	Df	2021-12-25	2021-12-18

Note: Source denotes the origin of the reference data; Number denotes the quantity from each source; Country signifies the country where the fire occurred; ID represents the identifiers obtained from EMSR and MTBS; Burned Area indicates the burned area in hectares (with MTBS providing acreage, converted uniformly to hectares herein);Climate Type indicates the climatic zones in which each wildfire event occurred; L8 Time and S2 Time respectively denote the image acquisition dates of Landsat-8 and Sentinel-2 downloaded for this study. The specific occurrence time of each wildfire can be checked through the URLs for EMSR and MBTS provided in the main text.

the respective study areas. Landsat-8 and Sentinel-2 provide medium-resolution remote sensing images, which are available for free and have been frequently used in previous studies on BA mapping [43], [44], [45], [46].

Herein, the Landsat-8 Collection 2 Level 2 T1 data with <10% cloud cover were downloaded from USGS (https://earthexplorer.usgs.gov/), specifically for bands 2–7. Sentinel-2 MSI2A datasets with <10% cloud cover, excluding bands 1, 9, and 10, were downloaded from the Copernicus data space ecosystem (https://dataspace.copernicus.eu/). The 2A designation for both datasets represented that they underwent geometric correction, radiometric calibration, and atmospheric correction [47]. Table I shows the specific wildfire information and the corresponding image acquisition dates used herein. All the acquired bands of Sentinel-2 were resampled to a resolution of 20 m. Based on the aforementioned explanations regarding image levels, cloud cover control, and resampling, the data used in this study has undergone rigorous quality control.

2) BA Reference Mask Data: To improve the efficiency and accuracy of creating BA labels, two types of BA products were used for manual labeling: EMSR provided by CEMS (https://emergency.copernicus.eu) and MTBS project (https://www.mtbs.gov/). As part of the European Union's Copernicus program, CEMS offers free forest BA maps using satellite imagery and other geographic data [37]. Most BA maps in EMSR are delineated from post-fire very high-resolution images, such as those from WorldView-2 and SPOT6/7, with approximately 0% cloud cover [32].

The MTBS project records the perimeters and burn severity of forest fires in the United States, covering fires of 1000 acres or more in the western U.S. and 500 acres or more in the eastern U.S. from 1984 to the present [42]. One of the greatest advantages of the MTBS project is the consistency of its data products, and the Landsat series of satellites is the primary data source. Since 2015, Sentinel-2 has been used for BA mapping to generate images with a spatial resolution of 30 m when high-quality Landsat data are unavailable via resampling [42].

Both types of reference mask data have been widely used in previous studies on BA mapping. Table I provides specific information about the reference BA mask data.

III. METHODOLOGY

A. Workflow

Fig. 2 shows the workflow of this study, which is divided into three main parts: data acquisition and preprocessing, model generation, and model performance. First, reference mask data were obtained, and Landsat-8 and Sentinel-2 images were downloaded based on BA location and time. The optimal band combinations for Landsat-8 and Sentinel-2 were determined based on the band selection principles. Using the reference mask data, labels were generated to create a DL database.

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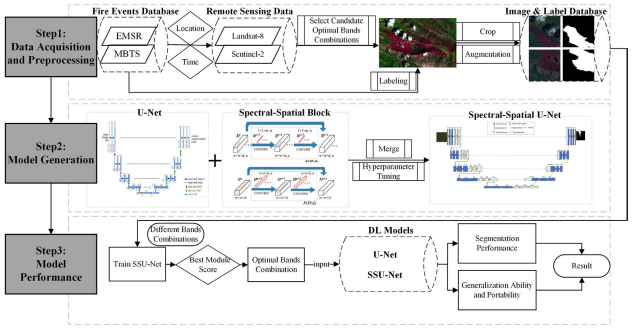


Fig. 2. Workflow of this study.

Subsequently, spectral–spatial blocks were integrated to enhance the performance of the U-Net model in generating the DL model tailored for this study. Finally, the segmentation accuracy and generalization ability of the new DL model were compared with those of U-Net.

B. Principles of Band Selection

In theory, the more spectral bands remote sensing imagery contains, the richer the information conveyed and the higher the classification accuracy [48]. However, by increasing the number of bands, redundancy may be introduced among them. This can potentially compromise the extraction of land cover information and lead to decreased accuracy [49]. Therefore, selecting an optimal band combination is crucial for effectively extracting remote sensing information and classifying remote sensing imagery. Four principles were established for selecting the optimal band combination based on spectral characteristics and model performance as follows:

1) *Information Content:* Each band should contain substantial characteristic information, and the combined bands should collectively provide a large amount of information, thereby facilitating feature extraction from remote sensing images.
2) *Correlation:* The correlation between bands should be minimal to avoid information redundancy that could negatively impact the extraction of land cover information.
3) *Class Separability:* The selected band combination should effectively distinguish the target land cover types. The optimal band combination should meet the requirements for distinguishing between BAs and unburned areas. The resulting composite color image should align with the theoretical understanding.
4) *Model Performance:* The selected band combination should exhibit the best performance and the strongest generalization ability in DL models.

Multiple metrics were used to address these four principles. For the first principle, two metrics were used: standard deviation and the optimum index factor (OIF) [50]. Standard deviation is used to measure the amount of information contained in a band. A larger standard deviation indicates greater contrast in the image, implying richer land cover information [51]. The standard deviation is calculated as follows:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \hat{\mu}\right)^2}. \tag{1}$$

In the equation

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2}$$

where n represents the total number of pixels in an image, x_i represents the reflectance value of the ith pixel, and $\hat{\mu}$ represents the mean reflectance value of the image.

OIF was proposed by Chavez Pat et al. [50] and is calculated using standard deviation and correlation coefficients. A higher OIF indicates that the band combination contains more information. OIF is expressed as follows:

$$\mathrm{OIF} = \frac{\sum_{i=1}^{3} S_i}{\sum_{j=1}^{3}\left|R_{ij}\right|}, \quad (j > i) \tag{3}$$

where S_i represents the standard deviation of the ith band and R_{ij} represents the correlation coefficient between the ith and jth bands.

For multiple bands, the correlation coefficient can indicate the degree of similarity between different bands. A higher correlation coefficient suggests greater redundancy between the bands. It is expressed as follows [52]:

$$\rho = \frac{S_{ij}}{\sqrt{S_{ii} \times S_{jj}}}. \tag{4}$$

In the equation

$$S_{ij} = \mathrm{Cov}(i, j) = \frac{1}{n}\sum_{k=1}^{n}\left(X_{ik} - \overline{X}_i\right) \times \left(Y_{jk} - \overline{Y}_j\right) \tag{5}$$

$$S_{ii} = \frac{1}{n}\sum_{k=1}^{n}\left(X_{ik} - \overline{X}_i\right)^2 \tag{6}$$

$$S_{jj} = \frac{1}{n}\sum_{k=1}^{n}\left(Y_{jk} - \overline{Y}_j\right)^2 \tag{7}$$

where i and j represent the number of bands and S_{ij} denotes the covariance between bands i and j. $\overline{X}_i$ and $\overline{Y}_j$ denote the average spectral reflectance values of bands i and j, respectively. X_{ik} and Y_{ik} represent the reflectance values of the kth pixel in the corresponding bands.

The points in BAs and nonburned forest areas in the images were uniformly selected and used for plotting spectral curves for different bands. Then, the identified bands, where significant differences in reflectance between BAs and unburned areas were observed. This process helped determine candidate optimal band combinations. Simultaneously, these bands were combined to achieve a display effect that aligned with the theoretical expectations such as showing BA in red, vegetation areas in green, and smoke or clouds in white.

After performing these steps, several candidate optimal band combinations were identified, which were individually inputted into the DL model. Based on the best model scores achieved by different band combinations, the final optimal band combination is determined.

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TABLE II
DETAILS OF DATA AUGMENTATION

Type	Method	Object
Geometric transformations	Rotation (90°; 180° and 270°)	Images and labels
	Flipping	Images and labels
Radiometric adjustment	Blurring	Images
	Illumination adjustment	Images
Adding noise	Adding Gaussian noise and salt-and-pepper noise	Images

C. Data Cropping and Data Augmentation

Remote sensing images need to be cropped before they are input into DL models to reduce processing time and increase data volume [53]. Based on De Bem et al.'s [54] conclusion on DL models using a window size of 256 × 256 pixels, Landsat-8 and Sentinel-2 images were cropped into multiple 256 × 256 patches using a sliding window approach. The cropping overlap rate was set to 0.25 [60].

Then, the data were split into training, validation, and test sets in an 8:1:1 ratio. Data augmentation was implemented to prevent model overfitting, increase the amount of usable training data, and enhance the model's robustness to data variations [56]. Geometric transformations [60], radiometric adjustment, and adding noise data augmentation techniques were used herein. The specific methods are detailed in Table II.

Using data augmentation, the number of images in the training, validation, and test sets was increased to ten times the original while maintaining the same proportion; thus, the final DL dataset was obtained.

D. Improved U-Net Generation

U-Net was improved to enhance its applicability for BA detection and improve the generalization between Landsat-8 and Sentinel-2 data. U-Net proposed by Ronneberger et al. [28] incorporated both long and short skip connections, linking feature maps from corresponding levels in the encoder and decoder. This framework effectively combined low-level detailed features with high-level semantic representations, showing promising performance in image segmentation. To enhance the classification accuracy and generalization ability of U-Net, an attention mechanism, namely the spectral–spatial module, was integrated into it herein.

This module was initially applied for hyperspectral image classification [55], the specific structure can be seen in Fig. 3. The spectral and spatial modules comprise 3-D convolution residual modules, each containing two convolution layers and batch normalization layers. In the spectral module, 24 convolution kernels of size 1 × 1 × 7 are used for spectral feature extraction. In the spectral module, skip connections are employed to prevent accuracy degradation during regularization convolution. In the spectral module, a padding strategy is applied to maintain the spatial size of the 3-D feature cube during convolution. Its output serves as the input to the spatial module. The spatial module also uses 24 convolution kernels of size 3 × 3 × 128, with different kernel sizes.

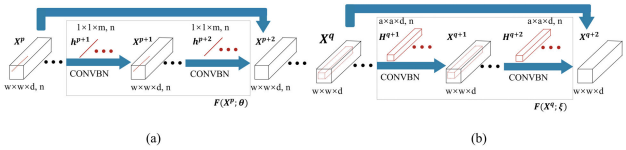


Fig. 3. Schematic of spectral–spatial modules (from [52]). (a) Spectral module and (b) spatial module.

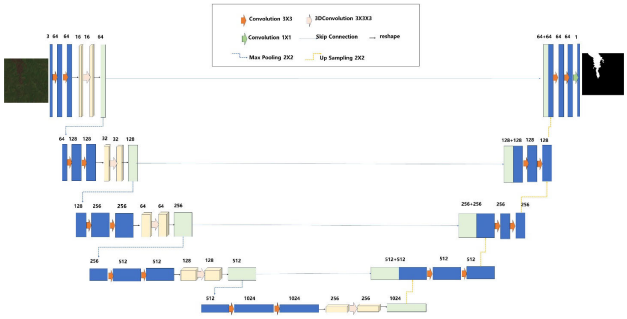


Fig. 4. Schematic of SSU-Net (adapted from [28]).

U-Net integrated with spectral–spatial module is referred to as the spectral–spatial U-Net (SSU-Net). The schematic of SS-Net is shown in Fig. 4.

SSU-Net included four down-sampling and four upsampling stages. In the first plane layer, the input image size is 256 × 256 × 3 and two 3 × 3 convolution kernels are used for 2-D convolution, which generate feature planes with 64 output channels. These planes are then reshaped into a 3-D cube and input into the spectral–spatial module. The spectral–spatial module was simplified to include two convolution layers and one normalization layer herein, each using 3 × 3 × 3 kernels and the LeakyReLU activation function. The feature cube outputted by this module was reshaped into a 2-D feature plane with 64 channels and further downsampled. The output from this layer was connected through a skip connection to the upsampled result. This process is repeated for each subsequent plane layer. During upsampling, 2 × 2 convolution kernels are used, and the decoded results are concatenated with the outputs from the corresponding levels of the encoder. This concatenated output is then processed with two 2-D convolution using 3 × 3 convolution kernels to reduce the number of channels in the feature map before proceeding with further upsampling. The final output of the upsampling process is a feature map with 64 channels, which is then converted into the final grayscale image using a 1 × 1 convolution.

It used the Adam optimizer with a batch size of 4 and a learning rate of 0.0001, the number of Epochs = 50. The environmental configuration of SSU-Net is detailed in Table III.

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TABLE III
DETAILS OF ENVIRONMENTAL CONFIGURATION

Experimental Environment	Details
CPU	13th Gen Intel® Core(TM) i9-13900KF
GPU	NVIDA GeForce RTX 4060
Operating system	Windows
Programming language	Python 3.8
Deep learning framework	Pytorch 2.0.1
Acceleration tool	CUDA 11.8

E. Accuracy Assessment

Accuracy assessment is crucial in BA mapping. Visual analysis methods can only perceive significant differences in case of substantial deviations in results. Therefore, evaluating model results using standard metrics is a more feasible approach to verify the effectiveness of the model. Herein, various common metrics were used for evaluating the effectiveness of the semantic segmentation model, including accuracy (ACC), intersection over union (IoU), recall, precision, and $F1$ score.

These metrics are expressed in (8)–(12). ACC represents the proportion of correctly classified pixels to the total pixels. IoU measures the overlap between the predicted and true BA pixels, ranging from 0 to 1. $F1$ score is a metric that combines both recall and precision. Recall measures the proportion of correctly predicted BA pixels to all actual BA pixels, whereas precision measures the proportion of correctly predicted BA pixels to all pixels predicted as BAs. The optimal model score mentioned in Section III-B refers to the highest $F1$ score achieved during training with different band combinations

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{8}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{11}$$

$$F1\ \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}} \tag{12}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

IV. RESULTS

A. Optimal Band Combination

1) Selection Based on Information Content and Correlation: Herein, single-band information, multiband information, and model performance were considered to determine the optimal band combination. First, the information of each band was evaluated based on its standard deviation. Fig. 5(a) shows the standard deviations of Landsat-8 bands, with band 6 exhibiting the highest value, followed by bands 5 and 7. In the visible light region of Landsat-8, the standard deviations in bands 2, 3, and 4 increased with longer wavelengths. Fig. 5(b) shows the standard deviations of Sentinel-2 bands, with a similar trend in the visible light region as Landsat-8. Sentinel-2 comprises three red edge bands (5, 6, and 7) that follow a consistent trend with the visible light region. Among the two NIR bands (bands 8 and 8A) of Sentinel-2, band 8 has a higher standard deviation than band 8A. Moreover, Band 12 had a larger standard deviation than band 11, which were the two SWIR bands.

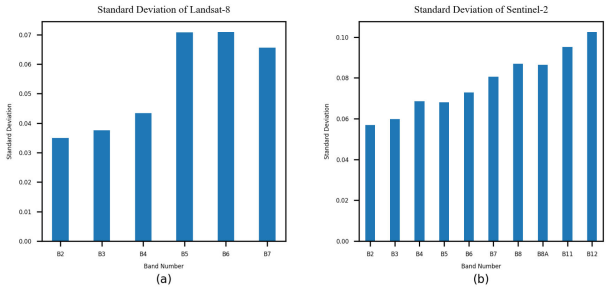


Fig. 5. Standard deviations of Landsat-8 and Sentinel-2 bands. (a) Landsat-8 bands and (b) Sentinel-2 bands.

Highly correlated bands should not be combined when considering multiband information, for which a correlation analysis must be performed. Fig. 6(a) shows the correlation coefficients among the six bands of Landsat-8. The correlation coefficient of band 5 with other bands was around 0.5, whereas that with band 7 was 0.18, indicating that band 5 has the highest independence and should be included in the band combinations. Considering the characteristics of Landsat-8 data, one band from the visible light bands (2, 3, and 4) was selected along with SWIR bands (6 and 7). Thus, six pseudo-color compositions were selected to calculate the OIF.

Fig. 6(b) shows the correlation coefficients among the ten bands of Sentinel-2. Similar to Landsat-8, one of the visible light bands (2, 3, and 4), NIR bands (8 and 8A), and SWIR bands (11 and 12) were selected for combination. Unlike those of Landsat-8, the red edge bands (5, 6, and 7) of Sentinel-2 have additional constraints: band 5 cannot be combined with other bands and bands 6 and 7 cannot be combined with NIR bands due to their high correlation coefficients. Therefore, 24 pseudo-color compositions were selected to calculate the OIF. The combinations of Landsat-8 and Sentinel-2 bands along with their corresponding OIFs are detailed in Table IV.

Table IV shows considerable differences between the two SWIR bands in different band combinations of Landsat-8. Although band 6 has a higher standard deviation than band 7, the combinations involving band 7 consistently outperform those of band 6 under similar conditions. Additionally, the OIFs of the three band 7 combinations show minimal differences. In different band combinations of Sentinel-2, the SWIR band shows a consistently higher OIF than band 12, indicating

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TABLE IV
COMBINATIONS OF LANDSAT-8 AND SENTINEL-2 BANDS ALONG WITH THEIR CORRESPONDING OIF

Source	Band Combination	OIF	Ranking	Source	Band Combination	OIF	Ranking
Landsat-8	Band 2-5-7	0.1404	1	Sentinel-2	Band 4-8-12	0.1550	1
	Band 4-5-7	0.1280	2		Band 4-8A-12	0.1548	2
	Band 3-5-7	0.1253	3		Band 2-8-12	0.1541	3
	Band 2-5-6	0.1129	4		Band 2-8A-12	0.1539	4
	Band 4-5-6	0.1035	5		Band 4-7-12	0.1473	5
	Band 3-5-6	0.0974	6		Band 3-8A-12	0.1472	6
					Band 3-8-12	0.1471	7
					Band 2-7-12	0.1457	8
					Band 3-7-12	0.1398	9
					Band 2-8-11	0.1284	10
					Band 2-8A-11	0.1273	11
					Band 4-8-11	0.1272	12
					Band 4-8A-11	0.1262	13
					Band 4-6-12	0.1232	14
					Band 2-7-11	0.1219	15
					Band 4-7-11	0.1214	16
					Band 2-6-12	0.1211	17
					Band 3-8-11	0.1208	18
					Band 3-8A-11	0.1201	19
					Band 3-6-12	0.1174	20
					Band 3-7-11	0.1152	21
					Band 4-6-11	0.1048	22
					Band 2-6-11	0.1044	23
					Band 3-6-11	0.0998	24

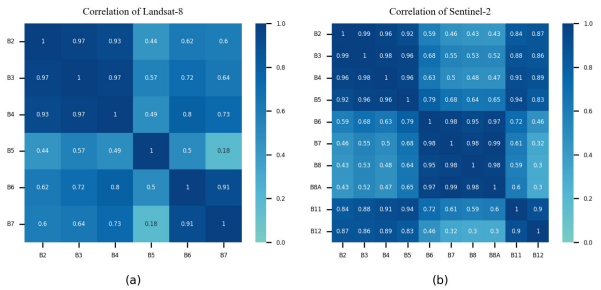


Fig. 6. Correlation coefficients among the bands of Landsat-8 and Sentinel-2. (a) Landsat-8 data and (b) Sentinel-2 data.

that band 12 conveys more information compared to band 11. Regarding the NIR bands, bands 7, 8, and 8A show minimal differences, suggesting that the differences in bands between the visible and NIR regions are small. Therefore, a more detailed band selection process must be undertaken.

2) Based on Class Separability: For class separability, BAs and unburned areas in forests were primarily distinguished; this difference was represented by the spectral curves of these two regions. Fig. 7(a) and (b) show the spectral curves of land cover for Landsat-8 and Sentinel-2, respectively. Red and green curves indicate BAs and unburned areas, respectively. Both images show distinct spectral characteristics for BAs, with considerably lower reflectance in the NIR bands than the unburned areas and higher reflectance in the SWIR bands. In Landsat-8, the difference in band 7 is greater than that of band 6, indicating higher sensitivity, whereas the differences between visible light bands are less pronounced and their values are similar. This indicates that a more comprehensive band selection process is required for the visible light bands of Landsat-8. Among the six OIF combinations of Landsat-8, band 6 was excluded from consideration and the remaining combinations underwent further selection process.

For Sentinel-2, the SWIR bands were easier to choose, as band 12 exhibited considerable differences from band 11. Band 6 did not show as much difference as band 7, whereas the differences of band 7 were very close to those of the two NIR bands, 8 and 8A, largely due to their similar wavelengths and the similar land cover information they contain. Additionally, the visible light bands were difficult to distinguish. Therefore, among the 24 combinations, those containing bands 11 and 6 were excluded and the remaining combinations were subjected to the next selection step.

3) Selection Based on Model Performance: Based on the three aforementioned principles, three band combinations were chosen for Landsat-8, namely bands 2-5-7, 3-5-7, and 4-5-7, and nine band combinations were chosen for Sentinel-2, namely bands 2-7-12, 3-7-12, 4-7-12, 2-8-12, 3-8-12, 4-8-12, 2-8A-12, 3-8A-12, and 4-8A-12 for data preparation to proceed with the selection based on the fourth principle. Then, the best display effect on these band combinations was deter-

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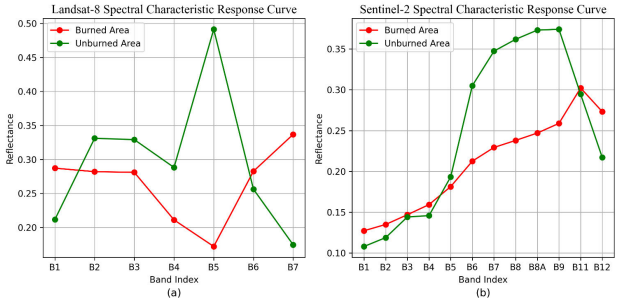


Fig. 7. Spectral curves of land cover for Landsat-8 and Sentinel-2. (a) Landsat-8 data and (b) Sentinel-2 data.

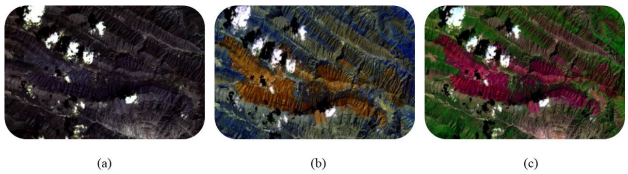


Fig. 8. Display effects of different band combinations for Landsat-8. (a) B4-3-2, (b) B7-6-5, and (c) B7-5-4.

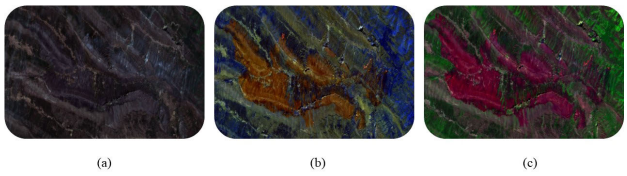


Fig. 9. Display effects of different band combinations for Sentinel-2. (a) B4-3-2, (b) B12-11-8, and (c) B12-8-4.

mined, which indicated that the SWIR, NIR, and visible light bands served as red, green, and blue channels, respectively, aligning with the theoretical findings. Figs. 8 and 9 show the comparative images. Figs. 8(a) and 9(a) show the true-color bands of the two datasets, along with the actual colors of the land cover. Figs. 8(b) and 9(b), based on the studies of Hu et al. [32] and Knopp et al. [34], show the display effect using Landsat-8 bands 7, 6, and 5 and Sentinel-2 bands 12, 11 as red, green, and blue channels, respectively. In these images, BA appears yellow-brown, vegetation appears blue-black, and bare soil appears black-brown, deviating from the theoretical expectations. In the images obtained using the arrangements determined herein [Figs. 8(c) and 9(c)], i.e., Landsat-8 bands 4-5-7 and Sentinel-2 bands 4-8-12, BA appeared red and vegetation appeared green, consistent with the theoretical expectations.

After determining the optimal display channels, various band combinations of Landsat-8 and Sentinel-2 were processed using sliding window cropping and data augmentation. These were then input into SSU-Net. Fig. 10(a) compares the accuracy of SSU-Net for the three band combinations of Landsat-8. The highest model score was achieved using the combination of bands 7-5-4. Fig. 10(b) compares the best model scores for the nine-band combinations of Sentinel-2, where the combination of bands 12-8-4 achieved the highest model score. Therefore, the optimal band combinations for Landsat-8 and Sentinel-2 were bands 7-5-4 and bands 12-8-4, respectively, for identifying BAs.

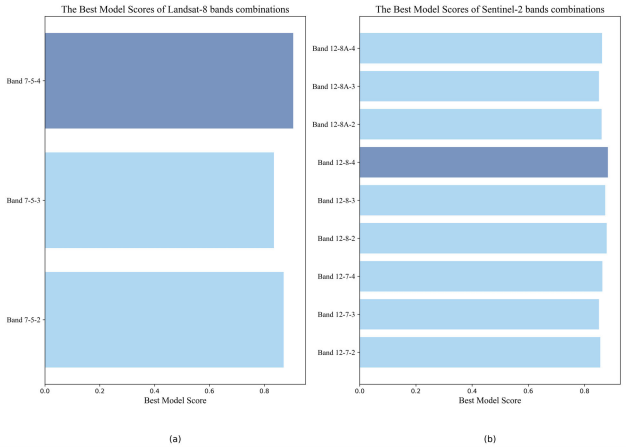


Fig. 10. Comparison of the best model score for the selected Landsat-8 and Sentinel-2 band combinations. (a) Landsat-8 data and (b) Sentinel-2 data.

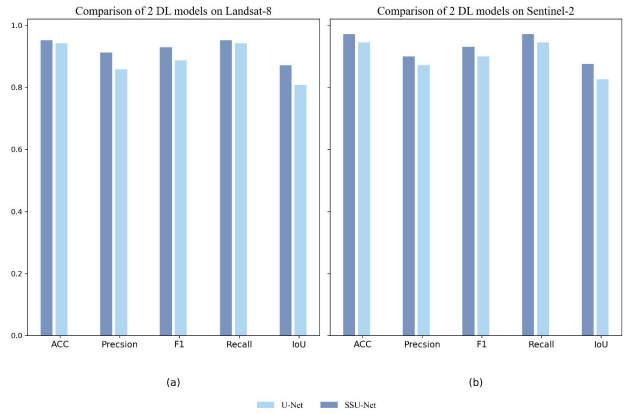


Fig. 11. Comparison of SSU-Net and U-Net on two datasets. (a) Landsat-8 data and (b) Sentinel-2 data.

B. Model Performance

1) Comparison of Classification Accuracy: The BA mapping abilities of SSU-Net and U-Net were compared based on Landsat-8 and Sentinel-2 data, as shown in Fig. 11. On both the Landsat-8 and Sentinel-2 datasets, SSU-Net outperformed U-Net across various accuracy metrics. On the Landsat-8 dataset, the $F1$ score and IoU of SSU-Net improved by 4.8% and 7.7%, respectively, compared with those of U-Net. On the Sentinel-2 dataset, the $F1$ score and IoU of SSU-Net improved by 3.4% and 5.9%, respectively, compared with those of U-Net. The images in columns of Fig. 12(c) and (d) provide two more intuitive examples for comparison. Fig. 12(c) shows the results from the U-Net for Landsat-8 and Sentinel-2, while Fig. 12(d) displays the results from SSU-Net. The shapes of the predicted region by the two methods are quite similar; however, SSU-Net performs better in terms of detail, identifying smaller BA in the images and avoiding misclassifications associated with U-Net's RGB-based processing, as seen in the second and fourth rows. SSU-Net also provides precise classification of pixels in areas outside the BA. This indicates that SSU-Net can more accurately identify and map BAs.

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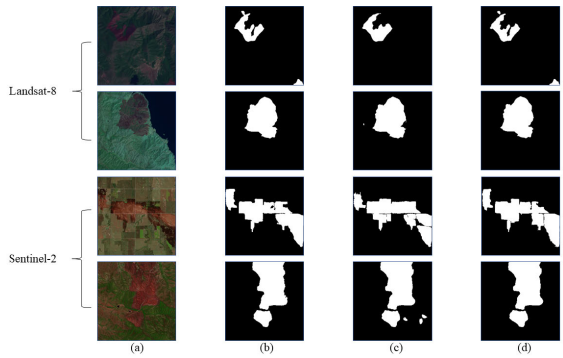


Fig. 12. Examples of segmentation results with the U-Net and SSU-Net, respectively, on the satellite dataset. (a) Image. (b) Label. (c) U-Net. (d) SSU-Net.

TABLE V
COMPARISON OF THE GENERALIZATION ABILITIES OF SSU-NET AND U-NET

Training Data	Test Data	Modules	Best Module Scores	Test Results
Landsat-8	Sentinel-2	SSU-Net	0.9334	0.9151
		U-Net	0.9155	0.8957
Sentinel-2	Landsat-8	SSU-Net	0.9093	0.9313
		U-Net	0.8797	0.8858

2) Generalization Ability of SSU-Net: The generalization ability of SSU-Net between datasets with spectral consistency was further evaluated. The model was trained on one data and tested on another. Then, the best model scores as well as test set results of SSU-Net and U-Net were compared, as shown in Table V. When training on Landsat-8 and testing on Sentinel-2 or training on Sentinel-2 and testing on Landsat-8, SSU-Net showed an outstanding performance. Under the former condition, the performance of SSU-Net improved by 2.1% compared with that of U-Net. Under the latter condition, the performance of SSU-Net improved by 5.1%, indicating its better generalization ability.

V. DISCUSSION

Herein, a DL model was developed for mapping BAs using multisource multispectral satellite imagery. To this end, Landsat-8 and Sentinel-2 were selected as the image data sources owing to their highly similar spectral band information and comparable spatial resolutions; these factors together enable near-real-time monitoring of Earth's surface. Recognizing the spectral consistency between Landsat-8 and Sentinel-2 data, optimal band combination was innovatively selected for detecting BAs based on their spectral characteristics, guided by four principles: information content, correlation, class separability, and model performance. Figs. 7 and 8 show that the selection of the optimal band combination positively contributed to the identification and extraction of BAs.

Band selection is a common approach in multispectral imagery used for enhancing experimental results by excluding less impactful bands [35], [56]. However, the existing methods typically use stepwise addition or subtraction of bands to compare the accuracy metrics such as $F1$ score and IoU of each band combination and define the combination with the highest accuracy as optimal. However, it is important to recognize that DL models are "black boxes" that rely heavily on their architectural settings rather than the inherent spectral characteristics of remote sensing imagery [57]. This introduces variability in results. In contrast, the four-point principle proposed herein for determining the optimal band combination considers the spectral characteristics of multispectral imagery while demanding superior model performance. This ensures the "optimality" of band combination in both spectral features and model effectiveness. Compared with those reported by Hu et al. [32] and Knopp et al. [34], the optimal band combination for Landsat-8 and Sentinel-2 used herein for BA mapping aligned well with the theoretical expectations. This highlighted that spectral-based band selection is a more comprehensive approach for mapping BAs in forests.

In recent years, 3-D convolution or combining 2-D and 3-D convolutions is a common approach used in the medical field [57], [58], [59]. This is because 3-D convolution can handle the three dimensions of images (width, height, and depth) simultaneously, allowing it to capture spatial information comprehensively. This ability is crucial in medical imaging, such as magnetic resonance imaging and computed tomography scans that contain important contextual information. Using only 2-D slices would result in the loss of this contextual information. However, this approach is not widely explored in the field of DL for mapping BAs based on remote sensing images. Herein, a 3-D spectral–spatial convolution module was integrated into the U-Net framework; the new model was referred to as SSU-Net. The traditional U-Net model comprises a series of down-sampling (encoding) and upsampling (decoding) convolutional layers each with three channels, primarily designed for processing 2-D image inputs (e.g., RGB images). In contrast to studies by Hu et al. [32] and Knopp et al. [34], a spectral–spatial module that transforms 2-D feature maps into 3-D feature maps and performs 3-D convolutions in the spectral–spatial domain was introduced herein. This approach enabled the extraction of more comprehensive spatial and spectral information, which enhanced the feature representation ability of the model. Such information may not be fully captured in the traditional 2-D plane, thereby improving the performance of SSU-Net.

The integration of the spectral–spatial module enhanced the segmentation accuracy of SSU-Net and improved its generalization ability over U-Net. To compare their performances, the experimental design approach reported by Hu et al. [32] was followed, wherein the performance of models was analyzed by training them on Sentinel-2 data and testing on Landsat-8 data and vice versa. Table V shows the results of generalization capability, demonstrating that SSU-Net outperforms U-Net under both conditions. Consistent with the findings of Hu et al. [32] models trained on Sentinel-2 data performed well on Landsat-8 data.

Drawing BA maps from post-fire single-date imagery is also challenging. SSU-Net has achieved satisfactory performance, but the study still has limitations. Due to constraints

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in reference data availability, the fire events selected herein were limited to Europe, North America, and South America. The impact of spatial heterogeneity in forest fires—such as differences in vegetation due to regional and climatic variations—on model accuracy is not yet clear. Furthermore, although SSU-Net can theoretically identify BAs from spectral features and should not misclassify mountain shadows or clouds as BAs, we utilized L2-level imagery without further exploring its performance on L1-level imagery in the study. The ratio of different datasets affects the model's learning capability, and the influence of varying dataset proportions on model performance should be evaluated. Regarding DL models, one of the focuses in this study is whether the spectral–spatial module in SSU-Net can enhance U-Net's ability to identify BAs, which requires comparison under identical hyperparameter conditions to be meaningful. This implies that there is room for improvement in the SSU-Net, as the default parameters of U-Net may not be optimal for SSU-Net. Despite the notable generalization and transferability of SSU-Net, differences in sensors among different data sources are significant factors that cannot be overlooked.

Future research should focus on the following aspects. Despite the promising potential demonstrated by DL models for mapping BAs, remote sensing imagery remains critical. Mapping between images with different spatial and temporal resolutions or with spectral heterogeneity is worth considering to create larger datasets in globally diverse fire-prone regions. In the future, we will explore model adaptability across different geographic regions and even globally. The response of DL algorithms to spatial heterogeneity in forest fires is also a key area of interest. Besides spectral features, further investigation into various hybrid methods in remote sensing and DL is needed to achieve results that satisfy researchers from different fields, such as exploring multisource data and multimodal fusion techniques.

VI. Conclusion

The automatic extraction of BAs from remote sensing imagery is a focal point and challenge in wildfire research. This study contributes by leveraging multisource remote sensing data to mitigate the difficulty of satellite revisit periods in obtaining post-fire images. Moreover, optimal band combinations were identified for detecting BAs using Landsat-8 and Sentinel-2 data, which reduced the complexity of visual interpretation and enhanced the accuracy of automated algorithms. Furthermore, the integration of a spectral–spatial module into U-Net significantly enhanced its sensitivity to spectral information in images. This enhancement is crucial for improving the model's accuracy in identifying BAs in remote sensing imagery. Additionally, our study validated the models' ability to effectively transfer knowledge between Landsat-8 and Sentinel-2 datasets. This validation is important as it demonstrates the robustness and adaptability of our approach across different satellite platforms, which is essential for consistent and reliable BA detection.

First, spectral analysis of the commonly used satellite remote sensing data sources, Landsat-8 and Sentinel-2, was performed. Considering single-band information, multiband information, and model performance, optimal band combinations were identified for detecting BAs: Landsat-8 using bands 7-5-4 and Sentinel-2 using bands 12-8-4. These band combinations have been experimentally proven to perform excellently in detecting BAs, providing robust support and guidance for subsequent remote sensing research.

Additionally, SSU-Net was developed based on spectral considerations within the U-Net framework. A series of experiments was conducted to evaluate the classification capability, transferability, and generalization ability of the SSU-Net model compared with other DL models. The results demonstrated that SSU-Net outperformed other models in effectiveness.

Moving forward, these findings underscore the potential for further advancements in remote sensing techniques for wildfire research. Future research directions may explore refining the spectral–spatial module, optimizing transfer learning strategies between diverse datasets, and expanding the applicability of DL models in broader environmental monitoring and disaster management contexts.

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Shuai Cui received the B.S. degree in forestry from Northeast Forestry University, Harbin, China, in 2022, where he is currently pursuing the Ph.D. degree.

His research interests include forest fire detection, assessment of wildfire severity, and post-fire vegetation recovery.



Jianyang Liu received the B.S. degree in geographic information science from Northeast Forestry University, Harbin, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include forest species diversity, forest parameter extraction, and the application of LiDAR in forest ecology.



Yuping Tian received the B.S. degree in geographic information science from Jilin Normal University, Siping, China, in 2018. She is currently pursuing the Ph.D. degree with Northeast Forestry University, Harbin, China.

Her research interests include forest disturbance target detection based on deep learning.



Ying Quan received the B.S. degree in geographic information science, the M.S. degree in forest management, and the Ph.D. degree in forestry from Northeast Forestry University, Harbin, China, in 2016, 2019, and 2023, respectively.

She is currently a Post-Doctoral Researcher with the College of Forestry, Northeast Forestry University. Her research interests include point cloud processing and remote sensing applications in forestry resources inventory.



Bin Wang received the B.S., M.S., and Ph.D. degrees in forestry from Northeast Forestry University, Harbin, China, in 2012, 2015, and 2019, respectively.

He is currently working in the discipline of forest management with Northeast Forestry University. His research interests include forest ecosystem carbon cycling.



Mingze Li received the Ph.D. degree in forestry from Northeast Forestry University, Harbin, China, in 2010.

He is currently a Full Professor with the College of Forestry, Northeast Forestry University. His research interests include ecosystem carbon cycling and forestry remote sensing.



Yuan Zhang received the B.S. degree in geographic information science from Northeast Forestry University, Harbin, China, in 2024, where she is currently pursuing the Ph.D. degree.

Her research interests include deep learning and the application of hyperspectral data in forestry.



Chengyuan Wang received the B.S. degree in geographic information science from Northeast Forestry University, Harbin, China, in 2024, where he is currently pursuing the M.S. degree.

His research interests include remote sensing in forestry, sustainable management of forest resources, and sustainable development assessment.



Yupeng Chang received the B.S. degree in geographic information science from Northeast Forestry University, Harbin, China, in 2024, where he is currently pursuing the M.S. degree.

His research interests include remote sensing in forestry and deep learning.

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